

Surjective isometries of the real $L^1[0, 1]$ ball are extremal

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Status. Candidate full solution, likely valid pending human review. The proof is for real L^1 . The complex case should be formulated separately using real supporting hyperplanes and phase functions.

Source problem. Bargetz, Dymond, and Pirk [1] prove that surjective isometries of the closed unit ball are extremal for many classical Banach spaces. They explicitly single out $L^1[0, 1]$ as the remaining classical case not covered by their Theorem 1.1 and write that they do not know whether surjective isometries of the closed unit ball of $L^1[0, 1]$ are extremal.

In condition (i) of Theorem 1.1 the notion of an almost exposed point is weaker than the standard notion of an exposed point, although we do not know whether the two notions may coincide. We postpone further details and discussion on this until Definition 3.6. For now, we note that all Banach spaces with the Radon-Nikodym property satisfy the condition (i) of Theorem 1.1. Consequently, Theorem 1.1 captures all ℓ_p and $L^p[0, 1]$ spaces with $1 < p < \infty$. We also capture ℓ_1 (via (i)) and c_0 , as well as all $C(K)$ spaces. Since ℓ_∞ and $L_\infty[0, 1]$ are isometrically isomorphic to $C(K)$ -spaces, the theorem applies to them as well. Hence, arguably the only ‘classical’ Banach space which evades Theorem 1.1 is $L^1[0, 1]$. We do not know whether surjective isometries of the closed unit ball of $L^1[0, 1]$ are extremal.

The source paper’s nonlinear proof under hypothesis (i) uses the closed convex hull of almost exposed points:

In Lemma 3.3, and hence in Proposition 3.2, we are able to show that on sets which are the closed convex hull of its extreme points, the identity is extremal among *linear* nonexpansive mappings. For a corresponding nonlinear result, we need the stronger condition that the underlying convex set is the closed convex hull of its almost exposed points.

Theorem 3.10 (Theorem 1.1, (i)). *Let X be a Banach space with the property that its unit ball $\mathbb{B}_X$ is the closed convex hull of its almost exposed points. Then surjective isometries are extremal in the space of nonexpansive self-mappings of the unit ball.*

The proof below does not use almost exposed points. It uses the full family of norming faces of the L^1 -ball instead.

Proof Intuition

Suppose

$$\text{id} = (1 - \lambda)F + \lambda G, \quad 0 < \lambda < 1,$$

where F and G are nonexpansive self-maps of the L^1 -unit ball. As observed in [1], F and G must themselves be isometries. The extra L^1 -specific observation is that every norming functional of a unit vector f must also norm $F(f)$ and $G(f)$. Hence $F(f)$ and $G(f)$ are trapped in the same norming face as f : they have the same sign as f and no mass outside the support of f .

This face preservation is very rigid when combined with isometry. A signed normalized indicator of a set A cannot be moved inside A : comparing distances to the signed normalized indicators of all subsets $B \subset A$ forces its image to put at least the normalized Lebesgue mass on every measurable subset of A , hence exactly that mass. The same argument then fixes signed simple unit vectors, and density fixes the whole unit sphere.

It remains to rule out movement inside the ball. Once the sphere is fixed, distances from an interior point to all signed normalized indicators are preserved. On small enough measurable pieces these distances read off the integrals of the positive and negative parts of the point. Atomlessness lets us make those pieces arbitrarily small, so the point itself is recovered.

Statement

Let (Ω, Σ, μ) be an atomless probability space and put $X = L^1(\mu; \mathbb{R})$. Let $\mathbb{B}_X$ be the closed unit ball of X , and let $\mathcal{M}$ be the convex set of all nonexpansive maps $\mathbb{B}_X \rightarrow \mathbb{B}_X$.

Theorem 1. *The identity map on $\mathbb{B}_X$ is an extreme point of $\mathcal{M}$. In particular, the identity map on the closed unit ball of real $L^1[0, 1]$ is extremal among nonexpansive self-maps.*

Corollary 1. *Every surjective isometry of the closed unit ball of real $L^1[0, 1]$ is extremal among nonexpansive self-maps of that ball.*

Proof. The corollary follows from the theorem and Lemma 3.1 of [1], which reduces extremality of surjective isometries of a convex body with nonempty interior to extremality of the identity by conjugating with the inverse isometry. Thus it suffices to prove the theorem. $\square$

Proof of the theorem

Assume

$$\text{id} = (1 - \lambda)F + \lambda G, \quad 0 < \lambda < 1,$$

with $F, G \in \mathcal{M}$. We prove $F = G = \text{id}$.

First, F and G are isometries. Indeed, for $x, y \in \mathbb{B}_X$,

$$\begin{aligned} \|x - y\|_1 &= \|(1 - \lambda)(F(x) - F(y)) + \lambda(G(x) - G(y))\|_1 \\ &\leq (1 - \lambda)\|F(x) - F(y)\|_1 + \lambda\|G(x) - G(y)\|_1 \\ &\leq \|x - y\|_1. \end{aligned}$$

Both inequalities must be equalities, so $\|F(x) - F(y)\|_1 = \|G(x) - G(y)\|_1 = \|x - y\|_1$.

Lemma 1 (Face preservation on the sphere). *Let T denote either F or G . If $f \in X$ and $\|f\|_1 = 1$, put*

$$A = \{\omega : f(\omega) \neq 0\}, \quad s = \operatorname{sgn} f.$$

Then

$$T(f) = sq$$

for some $q \geq 0$ supported in A with $\int_A q d\mu = 1$.

Proof. Let $\varphi \in L^\infty(\mu)$ be any norming functional for f , so $\|\varphi\|_\infty = 1$ and $\int \varphi f d\mu = 1$. Applying φ to the decomposition of f gives

$$1 = (1 - \lambda) \int \varphi F(f) d\mu + \lambda \int \varphi G(f) d\mu.$$

Both integrals are at most 1, since $F(f), G(f) \in \mathbb{B}_X$. Hence both are equal to 1. In particular, $T(f)$ lies in every supporting hyperplane of $\mathbb{B}_X$ at f .

The normalized supporting functionals at f are precisely those $\varphi \in L^\infty$ with $\|\varphi\|_\infty = 1$ and $\varphi = s$ a.e. on A . Taking the particular functional $s1_A$, equality $\int_A sT(f) d\mu = 1$ and $\|T(f)\|_1 = 1$ force

$$T(f) = 0 \text{ a.e. on } \Omega \setminus A, \quad sT(f) = |T(f)| \text{ a.e.}$$

Thus $T(f) = sq$, where $q \geq 0$, q is supported in A , and $\int_A q d\mu = 1$. □

Lemma 2 (The isometries fix the unit sphere). *Let T denote either F or G . Then $T(f) = f$ for every $f \in X$ with $\|f\|_1 = 1$.*

Proof. For a measurable A with $\mu(A) > 0$ and $\varepsilon \in \{-1, 1\}$, write

$$u_A^\varepsilon = \varepsilon \frac{1_A}{\mu(A)}.$$

By the previous lemma,

$$T(u_A^\varepsilon) = \varepsilon q_A$$

where $q_A \geq 0$, q_A is supported in A , and $\int_A q_A d\mu = 1$. If $B \subset A$ has positive measure, then

$$\|u_A^\varepsilon - u_B^\varepsilon\|_1 = 2 \left(1 - \frac{\mu(B)}{\mu(A)} \right).$$

Since T is an isometry and $T(u_B^\varepsilon)$ is supported in B with sign ε , we also have

$$\begin{aligned} 2 \left(1 - \frac{\mu(B)}{\mu(A)} \right) &= \|T(u_A^\varepsilon) - T(u_B^\varepsilon)\|_1 \\ &\geq 2 \left(1 - \int_B q_A d\mu \right). \end{aligned}$$

Thus $\int_B q_A d\mu \geq \mu(B)/\mu(A)$ for every $B \subset A$. Since both measures have total mass 1 on A , $q_A = \mu(A)^{-1}1_A$ a.e. Therefore T fixes every signed normalized indicator u_A^ε .

Now let f be a signed simple unit vector,

$$f = \sum_{i=1}^n \varepsilon_i \alpha_i \frac{1_{A_i}}{\mu(A_i)},$$

where the A_i are pairwise disjoint measurable sets of positive measure, $\varepsilon_i \in \{-1, 1\}$, $\alpha_i > 0$, and $\sum_i \alpha_i = 1$. By face preservation,

$$T(f) = sq, \quad s = \operatorname{sgn} f,$$

where $q \geq 0$, q is supported in $\bigcup_i A_i$, and $\int q = 1$. For $u_i = \varepsilon_i 1_{A_i} / \mu(A_i)$, already fixed by T ,

$$\|f - u_i\|_1 = 2(1 - \alpha_i).$$

The same lower-bound argument gives $\int_{A_i} q d\mu \geq \alpha_i$. Summing over i yields equality for every i . If $B \subset A_i$, then

$$\left\| f - \varepsilon_i \frac{1_B}{\mu(B)} \right\|_1 = 2 \left(1 - \alpha_i \frac{\mu(B)}{\mu(A_i)} \right),$$

and the lower-bound argument gives

$$\int_B q d\mu \geq \alpha_i \frac{\mu(B)}{\mu(A_i)}.$$

Since $\int_{A_i} q d\mu = \alpha_i$, it follows that $q = \alpha_i \mu(A_i)^{-1}$ a.e. on A_i . Hence $T(f) = f$.

Signed simple unit vectors are dense in the unit sphere of $L^1(\mu)$, and T is continuous because it is an isometry. Therefore T fixes the whole unit sphere. $\square$

Lemma 3 (Fixing the sphere fixes the ball). *Let $T : \mathbb{B}_X \rightarrow \mathbb{B}_X$ be an isometry that fixes every point of the unit sphere of $X = L^1(\mu; \mathbb{R})$, where μ is atomless. Then $T = \operatorname{id}$.*

Proof. First $T(0) = 0$. If $z = T(0) \neq 0$, choose a measurable set B of positive measure on which z has one sign and $\int_B |z| d\mu > 0$. Taking the oppositely signed normalized indicator u on B , which is fixed by T , we get

$$1 = \|0 - u\|_1 = \|T(0) - T(u)\|_1 = \|z - u\|_1 = 1 + \int_B |z| d\mu + \int_{\Omega \setminus B} |z| d\mu > 1,$$

a contradiction.

Let $x \in \mathbb{B}_X$ and set $z = T(x)$. Since $T(0) = 0$, $\|z\|_1 = \|x\|_1$. For every measurable B with $\mu(B) > 0$ and every $\varepsilon \in \{-1, 1\}$, the signed normalized indicator $u_B^\varepsilon = \varepsilon 1_B / \mu(B)$ is fixed. Therefore

$$\|x - u_B^\varepsilon\|_1 = \|z - u_B^\varepsilon\|_1.$$

For any $w \in L^1(\mu)$,

$$\|w - u_B^\varepsilon\|_1 = \|w\|_1 + 1 - 2 \int_B \min \left((\varepsilon w)_+, \frac{1}{\mu(B)} \right) d\mu.$$

Because $\|x\|_1 = \|z\|_1$, we get

$$\int_B \min \left((\varepsilon x)_+, \frac{1}{\mu(B)} \right) d\mu = \int_B \min \left((\varepsilon z)_+, \frac{1}{\mu(B)} \right) d\mu$$

for every B and every sign ε .

Fix ε and $M > 0$. Let

$$E_M = \{(\varepsilon x)_+ \leq M\} \cap \{(\varepsilon z)_+ \leq M\}.$$

If $C \subset E_M$, atomlessness and finiteness of μ allow us to partition C into finitely many measurable pieces C_j with $\mu(C_j) < 1/M$, ignoring null pieces. On each C_j the truncation level $1/\mu(C_j)$ is larger than M , so the displayed equality gives

$$\int_{C_j} (\varepsilon x)_+ d\mu = \int_{C_j} (\varepsilon z)_+ d\mu.$$

Summing over j gives equality of the integrals over C . Since this holds for every $C \subset E_M$, the two functions $(\varepsilon x)_+$ and $(\varepsilon z)_+$ agree a.e. on E_M . Letting $M \rightarrow \infty$, they agree a.e. on Ω . Applying this for $\varepsilon = 1$ and $\varepsilon = -1$ gives $x^+ = z^+$ and $x^- = z^-$. Hence $x = z$. $\square$

Applying the last two lemmas to F and G , both maps fix the unit sphere and hence the whole closed unit ball. Thus $F = G = \text{id}$, so the identity is extremal.

Verification and novelty check

The proof is analytic and self-contained except for two source-paper facts: the reduction from surjective isometries to the identity, and the elementary observation that summands in a convex decomposition of an isometry are isometries. Both are reproduced or cited above.

Local index search found no existing full packet for arXiv:2409.04292. The only exact run hit was the preceding partial packet `2409.04292_l1_almost_exposed_atoms`, now superseded by this candidate full solution. Web searches on 2026-06-19 for the exact $L^1[0, 1]$ extremality phrases and related “extreme nonexpansive mappings L1” terms found the source paper but no later answer. This is not a proof of novelty, but it supports treating the packet as a candidate new solution.

Human review should focus on the two rigidity lemmas: first, the domination argument for signed normalized indicators; second, the local recovery of interior points from distances to signed normalized indicators.

References

- [1] Christian Bargetz, Michael Dymond, and Katriin Pirk. On extremal nonexpansive mappings. arXiv:2409.04292, 2024. <https://arxiv.org/abs/2409.04292>.